

SQL-Based Dynamic Pricing Intelligence System

Project Plan & Proposal

Abhijeet Patil



Problem Statement

Static pricing cannot adapt

Causes revenue and sales loss

Manual decisions lack data insights

No automated pricing system



Project Objectives

01

Analyze product sales data

02

Identify demand patterns

03

Suggest pricing decisions

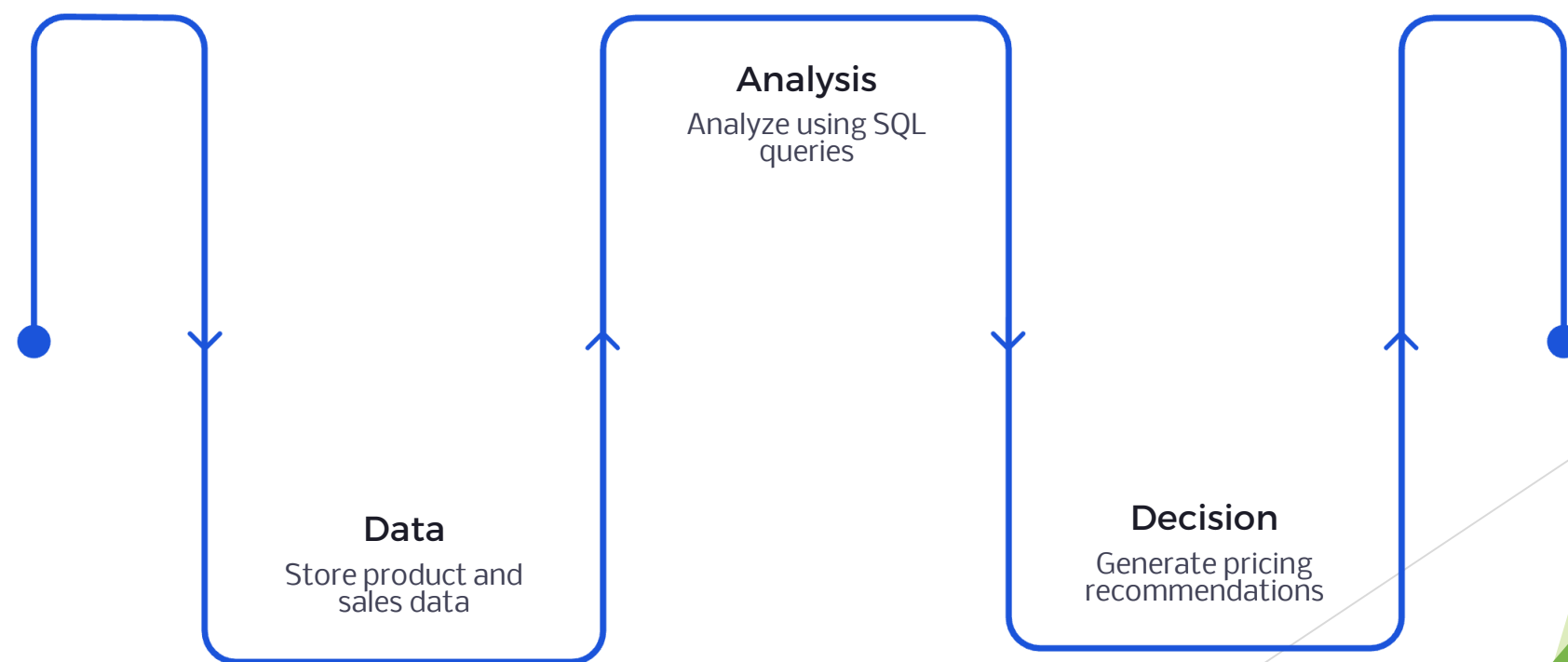
04

Increase, decrease, or maintain prices

Proposed Solution

How It Works

- Store product and sales data
- Analyze using SQL queries
- Generate pricing recommendations



System Architecture



Data Layer

Product and sales data storage



Analysis Layer

SQL processing and calculations



Decision Layer

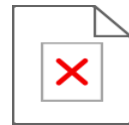
Pricing suggestions generated

SQL Concepts Applied



SELECT, WHERE

Data filtering operations



JOIN operations

Combine related tables



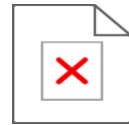
GROUP BY, SUM

Aggregate data analysis



CASE statements

Decision logic implementation



Date functions

Trend analysis over time

Database Design & Features

Table Structure

- Products table with ID, name, price
- Sales table with ID, product ID, quantity, date
- One-to-many relationship established



Demand Analysis

Pricing Engine

Revenue Insights

Trend Analysis

Implementation Methodology



Database Design



Data Insertion



SQL Development



Result Analysis



Expected Outcomes & Challenges

Expected Results

- ☐ Identify high demand products
- ☐ Generate pricing suggestions
- ☐ Provide sales insights

Challenges to Address

- ☐ Designing correct logic
- ☐ Writing efficient queries
- ☐ Handling data properly

Future Scope & Conclusion

Future Enhancements

Real-Time Integration

Live data streaming

Machine Learning

AI-powered pricing

Dashboard Visualization

Interactive UI interface

SQL enables data-driven pricing decisions

Solves real business problem with simple, scalable system